

MT-LNN: Microtubule-Inspired Liquid Neural Network

Shattering the Memory Wall for the Next Generation of Long-Context AI

Investor Pitch Deck

Awareness

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Website: <https://awareliquid.ai>

Paper: <https://huggingface.co/EverestAn/MT-LNN>

GitHub: <https://github.com/everest-an/M1>



- ▶ **1. Symbolism:** Logical and visual reasoning, closest to human conscious thought.
- ▶ **2. Bayesianism:** Prior/posterior probability reasoning via Bayesian networks.
- ▶ **3. Analogizer:** SVMs and nearest-neighbor classification.
- ▶ **4. Evolutionism:** Mutational and evolutionary algorithms.
- ▶ **5. Connectionism (Current Mainstream):** Neural networks, Large Models, XAI, and **Liquid Neural Networks (LNN)**.
- ▶ **6. Reinforcement Learning:** Reward-punishment optimization.
- ▶ **7. Multi-Agent Systems:** Simulating interaction between multiple intelligent agents.
- ▶ **8. Continual Learning:** AI evolving continuously without total retraining.



- ▶ **The Problem:** Large Language Models (LLMs) are hitting a physical “Memory Wall”. Scaling past 200K tokens leads to exponential costs and severe Out-Of-Memory (OOM) constraints during deployment.
- ▶ **The Solution:** MT-LNN replaces static Transformer mechanisms with a continuous-time Liquid Neural Network inspired by human brain microtubules.
- ▶ **The Edge:** Constant $O(1)$ memory footprint, eliminating the KV Cache dependency. High performance in ultra-long context reasoning ($> 128K$ tokens) on consumer-grade hardware.
- ▶ **The Ask:** Seeking strategic funding to scale our models from the 1.5B component level to a fully native 7B Enterprise architecture.

The AI Industry's Memory Crisis



- ▶ **The Cost of Long Context:** The AI industry is in an arms race for 200K+ token windows, but the underlying computational and memory costs are exploding.
- ▶ **OOM (Out-of-Memory) Limits:** For on-premise and local deployments, the true bottleneck isn't logic or processing power—it is strictly physical VRAM limits.
- ▶ **The Compute Trap:** Cloud providers are forced to stack phenomenally expensive GPU clusters to handle concurrent, long-context user requests due to legacy architecture flaws.

The Root Cause: KV Cache Bottleneck

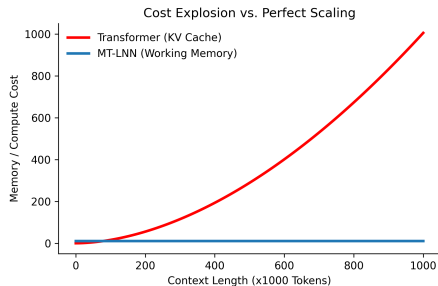


- ▶ **A Structural Limitation:** The standard Transformer architecture operates like an exhaustive recorder; predicting the next token requires holding all previous tokens' features in memory (KV Cache).
- ▶ **Complexity Disaster:** As context grows, memory complexity scales linearly $O(N)$, but compute complexity scales quadratically $O(N^2)$.
- ▶ **Physical Constraints:** Processing a 100,000-word document instantly fills an expensive A100 GPU. Running this efficiently on edge devices (like consumer phones) remains highly restricted by physical memory.

Cost Explosion vs. Perfect Scaling



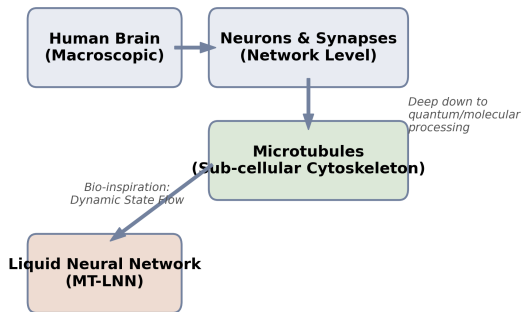
- ▶ Transformer family models depend on growing KV cache, leading to rising serving cost and memory pressure under long-context traffic.
- ▶ MT-LNN uses a compact recurrent latent state and microtubule-inspired selective dynamics to control memory growth.
- ▶ Product implication: lower deployment friction for private/on-prem and edge scenarios.



Biological Foundation: What are Microtubules?



- ▶ **Sub-cellular Processing:** While conventional AI mimics the network of neurons, actual biological cognition is regulated at a deeper level within the **Microtubules**—the cell's cytoskeleton.
- ▶ **Quantum-Level Dynamics:** Microtubules process information through continuous, dynamic protein states rather than binary synapses, enabling higher-dimensional fluid data integration.
- ▶ **The Pathway to LNN:** This biological flow directly inspires the **Liquid Neural Network (LNN)**, creating an architecture that inherently adapts to time-series data without rigid matrices.





- ▶ **Refusing to Store Every Pixel:** The human brain does not—and physically cannot—memorize the exact pixels of every scene spanning a decade.
- ▶ **Working Memory & Selective Forgetting:** Human cognition relies heavily on “Working Memory” (retaining core logic) and “Selective Forgetting” (discarding irrelevant noise).
- ▶ **AI’s Blind Spot:** Current LLMs are purely vast matrix multiplications lacking this dynamic time-flow mechanism or a Global Workspace Theory (GWT) path for conscious-level integration.

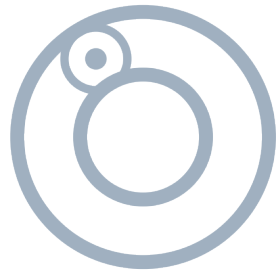
What is a Liquid Neural Network?



- ▶ **Dynamic Flowing States:** Liquid Neural Networks (LNNs) maintain an internal state that fluidly adapts to new data over time, contrasting sharply with static Transformer weights.
- ▶ **Compression and Abstraction:** The core mechanism: extract the essential synopsis, forget redundant filler, and compress thousands of words into a high-dimensional, time-evolving hidden state.
- ▶ **Brain-Like Foundation:** It is the first architecture to physically and dynamically mimic how microtubule structures process information in biological cells.



- ▶ **The Probabilistic Illusion:** Traditional Transformers can perfectly predict the sequence “*the apple fell from the tree to the ground.*” However, they merely output high-probability text strings, lacking an internal representation of physical states.
- ▶ **Causal Physical Perception:** MT-LNN breaks this limitation. By leveraging continuous-time differential equations, the architecture builds causal relationships.
- ▶ **Simulating Reality:** It dynamically evolves its internal hidden states, fundamentally grasping the physical process of the **apple falling** over time rather than just memorizing word frequencies.



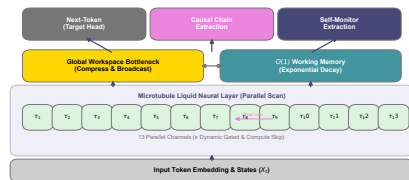
MT-LNN Dynamic Concept Positioning

The MT-LNN Architecture



Microtubule Liquid Neural Network:

- ▶ **Replacing the Cache Wall:** We completely discard the traditional $O(T)$ KV Cache, introducing an $O(1)$ **Working Memory Decay** matrix. Context size scales infinitely with zero added memory footprint.
- ▶ **Replacing Static FFNs:** The architecture integrates 13 parallel, continuous-time liquid network pathways equipped with *endogenous compute skipping*.
- ▶ **A Cognitive Stack, Not Just a Mixer:** a **GWT global-workspace bottleneck** (expert channels compete for a shared broadcast), **sleep-cycle consolidation** (NREM-style synthesis compresses short-term traces into durable structure), and a **fast-weight memory** that carries key→value bindings across windows and sessions.



Two Product Lines: Cognitive Slow Thinking + Edge Fast Thinking

M-series (M1/M2) — Cognitive Slow Thinking

- ▶ Hybrid: a frozen open-weight base (e.g. TinyLlama-1.1B) + our trained MT liquid adapter at $\sim 1\%$ parameter overhead.
- ▶ Ships **GWT workspace + sleep consolidation + EWC continual learning** on top of full base quality.
- ▶ Unique edge: **cross-window / cross-session associative recall** — 0.56 where frozen attention and LoRA are structurally **0.000**.

Both lines share one cognitive stack — GWT workspace, predictive coding, liquid time constants. Live demos: <https://awareliquid.ai>

O-series (O1) — Edge Fast Thinking

- ▶ Attention-free, trained from scratch (48M–125M), liquid ODE dynamics, millisecond latency.
- ▶ **Genuine $O(1)$ inference memory:** carried state flat at **0.381 MB** from 512 to 1,048,576 tokens, where a matched KV cache grows to **3,072 MB** ($8,063\times$).
- ▶ Targets: wearables, automotive, industrial streaming sensors.



- ▶ **Predictive Coding System:** Moving beyond pure “next word guessing”, MT-LNN introduces inter-scale predictive coding. Abstract reasoning channels natively self-supervise faster perception channels, forcing deep logic comprehension.
- ▶ **Endogenous Compute Skipping:** Utilizing dynamic κ logic gates, the architecture autonomously shuts off dormant channels mid-computation, yielding **exponential ROI cost-savings** on inference chips.
- ▶ **Explainable Causal Logging:** Say goodbye to black boxes. MT-LNN state matrices are natively parsed by *Causal Chain* and *Self-Monitor* extraction heads, rendering internal logical stepping completely observable.

Recent Upgrade 1: Spatial Computing — Letting the AI “Rehearse the World” in Its Mind



Beyond just “reading text”: it builds a mental map of the scene and, like a person, “thinks before it acts”:

- ▶ **Simulates how things unfold:** rolls the current situation forward a few steps in its mind — “what happens next”.
- ▶ **Judges plausibility:** spots when “something feels off”; implausible outcomes get flagged, suppressing **hallucination** at the source.
- ▶ **Physical intuition:** “nudge the cup and it tips”, “where will the ball land” — basic physical & spatial common sense.
- ▶ **Remembers where it has been:** records as it goes, then recalls the scene later from just an **approximate location**.

Value to customers:

- ▶ Upgrades the AI from “can talk” to “**can rehearse & can plan**” — the prerequisite for autonomous agents and robots.
- ▶ Rules out bad options in its head first: **less nonsense, fewer wrong actions**, more reliable decisions.
- ▶ Gets this “cognitive map” at **zero extra cost** on-device — plug-and-play.

Technical basis: a hippocampal-entorhinal-inspired spatial-cognition stack (localize → spatial memory →

Recent Upgrade 2: A Verifiable Reasoning Core — Every Step Is “On the Record, Never Quietly Drifting”



What does “verify” mean? A normal LLM is a black box — you can only “trust it”. We break the reasoning into independent **building blocks**, write each block’s required behaviour as a **math rule with a self-check**, and run them automatically on every build:

- ▶ **The machine checks its own work.** E.g. “energy can’t appear from nowhere”, “the direction math doesn’t drift on curved spaces”, “logic stays self-consistent” — like a **factory QA gate** for the AI.
- ▶ **Not “trust me” but “run it yourself”:** anyone can reproduce every claimed behaviour, line by line, with the same scripts.
- ▶ **We state exactly how sure we are:** the companion white paper honestly grades each claim into three tiers (rigorously proven / pinned by automated tests / empirically observed) — no

Value to customers:

- ▶ **Trust = the moat.** Finance, legal, medical and government fear “can’t explain why” most — every step here can be checked by a third party.
- ▶ Versus black-box LLMs, we give **behaviour guarantees**, not “probably fine”.
- ▶ **1232 automated tests** permanently green on CPU — every claim backed by a script and an artefact.

Recent Upgrade 3: Continual Learning — “Understands You More Over Time”, Without Forgetting the Old



LLMs have an old flaw: teach them something new and they often forget what they knew (the industry calls it “catastrophic forgetting”). We make the model **review the old while learning the new**, like a person:

- ▶ **Learns new, reviews old.** The model keeps a small “key-points notebook” (fixed size, no memory bloat) and interleaves old knowledge with new, so it learns without forgetting.
- ▶ **You can score how well it learns.** Three plain metrics quantify it — how much it forgot, how much it kept, whether it can generalize — no guessing.
- ▶ **There’s a ready demo:** on the same data stream, “with review” vs “without review”, side by side, with results pinned by automated tests.

Value to customers:

- ▶ **On-device lifelong personalization:** fits each user better the more it is used, with **data staying on the device** — no cloud.
- ▶ Directly attacks the pain of “fine-tune once and forget everything” — a **differentiated moat** for edge personalization.

Recent Upgrade 4: Autonomous Agent — It Already Runs the Full “See → Remember → Think → Decide → Act” Loop



We wired these capabilities into an agent that does the work itself, letting it solve a wayfinding task on its own. Every stage of the chain is a **real module** — **not a single faked step**:

- ▶ **See** (perceive where things are) → **Remember** (recall places it has been, even from a fuzzy impression) → **Think** (rehearse a few steps in its mind: “does this route work”) → **Decide** (lock onto the goal top-down; distractions are gated off) → **Act** (output a move) — the loop runs itself.
- ▶ **Value to customers:** this is exactly the core paradigm for **robots / agents / digital workers** — turning “perceive-remember-reason-decide” from a slide-deck vision into **code that runs**. (Currently an end-to-end integration-demo prototype, not yet a shipped product — we label it honestly.)

Recent Upgrade 5: From Capability to Product — A “Fast-Reflex Slow-Thinking” Sentry, Ready to Deploy



We built a deployable “sentry” product (DualSpeedSentry) that works much like a person:

- ▶ **Saves power normally, thinks hard only when it matters.** It watches with a cheap “reflex” day to day, and only **wakes the “deep thinking”** when a real anomaly appears — **big compute savings**, no idle spinning like legacy systems.
- ▶ **Doesn’t lose control when offline.** If the signal drops, it “coasts” on inertia for a bit or **stops safely** — never flails.
- ▶ **Output always stays within safe bounds.** The action magnitude is **capped by math**, guaranteeing no command beyond mechanical limits.

Ready to deploy:

- ▶ A serving interface (with acceptance tests) + container packaging & ops tooling — **live out of the box.**

Value to customers:

- ▶ A deployable **security / industrial-monitoring / defense** vertical: saves compute, stays safe, and ships immediately.



- ▶ **Hyper-Specialization:** We dedicate architecture to complex reasoning rather than storing vast arrays of encyclopedia trivia.
- ▶ **Laser-Focused Optimization:** Parameter capacity is heavily invested in *hardcore logical reasoning* and *massive cross-context structural memory constraints*.
- ▶ **Enterprise Application:** Perfect for delivering world-class semantic extraction on a 50,000-word financial audit or a massive legacy codebase branch, strictly operating on local secure hardware.

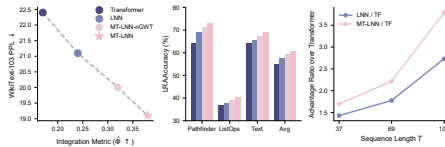
Core Performance 1: Selective Copy & Noise Resistance



Surviving Massive Noise ($T = 229$):

- ▶ **Length stress test:** As noise floods the context, every model's whole-sequence recall degrades. Under a fair full-sequence decode, the Transformer baseline drops to **10.9%** sequence-exact accuracy and LNN to **17.2%**.
- ▶ **MT-LNN edge widens:** By autonomously filtering what to retain via liquid states, MT-LNN holds the lead at **21.9%** – roughly **2 $\times$** the Transformer at this length.

At an ultra-compact scale ($\sim 200K$ parameters), MT-LNN leads on whole-sequence recall at every length – a modest $\sim 1.3\times$ margin over the Transformer at $T = 32$ (89.5% vs 67.6%) that widens toward $\sim 2\times$ as the stream grows.



Core Performance 2: Memory That Survives a Dropped Context Window



- ▶ **A task attention cannot do by construction:** store key→value facts, then *drop the entire KV cache / context window*, then ask again. Frozen attention and LoRA score **0.000** — the information is simply gone.
- ▶ **MT-LNN fast-weight memory: 0.56 mean recall** (3 seeds) across the dropped window. Ablate the fast-weight matrix and recall collapses to 0.008 — the state *is* the memory.
- ▶ **Cross-session persistence:** snapshotting the (F, z) state to disk and restoring it in a fresh process is **bit-exact lossless** — “remembers you across sessions” as an architectural property, not an external RAG database.
- ▶ **Honest boundary:** this is episodic key→value memory, not a long-context language-modeling gain (a measured null out-of-window) — we state exactly what it is and is not.

Validated Use Cases (Shipped or Benchmarked)



- ▶ **Battery management (NASA PCoE):** streaming cell state-of-health regression with a **2.6 KB** carried state; **+7.7% accuracy at 80% missing samples** — robustness to irregular sampling is native to liquid ODEs, not a patch.
- ▶ **Wearables — open-source wake word (O1-Sound):** always-on keyword spotting with a **5.12 KB constant state**; no cloud, millisecond-class latency on MCU-grade hardware.
- ▶ **Financial documents (M2 adapter):** **91.7% accuracy (44/48)** on document-extraction QA, fully on-prem.
- ▶ **Pipeline (scenario, not commitment):** automotive cabin systems, on-device assistants, industrial sentry robots (DualSpeedSentry serving stack already containerized).

Live demos and reproduction scripts: <https://awareliquid.ai> · GitHub `everest-an/M1`

The Ultimate Vision: Awareness Network

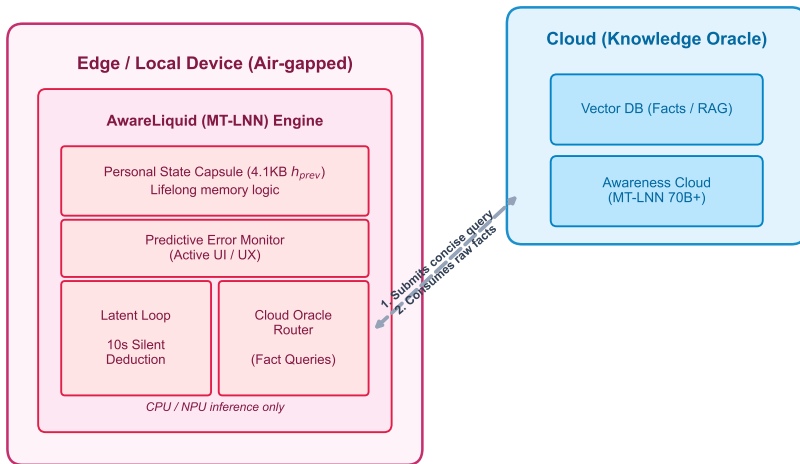
(Hybrid Edge-State Intelligence System)

Asymmetric Warfare vs. The Cloud Paradigm



- ▶ **Strategic Disruption:** Rather than relying on legacy Transformer APIs like Gemini and GPT-4o, we are vertically integrating our breakthrough MT-LNN architecture into an end-to-end ecosystem.
- ▶ **Core Philosophy:** The Edge **AwareLiquid Engine** serves as the continuous "soul" and reasoning engine, while our proprietary **Awareness Cloud** (High-Param MT-LNN) acts as the global knowledge Oracle.
- ▶ **Hybrid Edge-State Architecture:** The Awareness Network runs purely local logic natively, routing to the Awareness Cloud only for factual retrieval while maintaining absolute data sovereignty.
- ▶ **Privacy by Physics:** Core cognitive state remains perpetually contained within a local 4.1 KB $O(1)$ State Capsule, completely severing the "harvest and charge" cycle of legacy providers.

Awareness Network Architecture



The End of the "Token-Burning" Compute Arms Race



▶ The Legacy "Compute Trap":

- ▶ The Transformer paradigm forces every interaction to re-transmit and re-compute massive context histories, locking the industry into an endless "Token-Burning" death spiral.

▶ Our Paradigm: Brain-like Edge Engine + Cloud Knowledge Base

- ▶ **Local Brain (AwareLiquid):** 90% of logical reasoning, continuous memory, and private deliberation are handled purely on-device via a lightweight MT-LNN, achieving **Zero-Cost Local Compute**.
- ▶ **Awareness Cloud:** The cloud is no longer a high-frequency token black hole. It serves simply as a heavy-parameter knowledge base.

▶ The Business Reality: Escaping the Compute Wars

- ▶ Local compute resolves the vast majority of tasks. Ultra-lightweight knowledge retrieval is triggered *only* for factual blind spots—making scalable AI finally free from server-farm burn rates.

Four Pillars of the Experiential Shift



- ▶ **1. Latent Reasoning Loop (Silent Deep Deduction):**
 - ▶ **Cloud AI:** Fakes reasoning by spitting out verbose Chain-of-Thought (CoT) tokens.
 - ▶ **M1:** Embarks on 10 seconds of "silent calculation" running internal ODEs natively. It strikes truth without bloated text, exhibiting wise, human-like intuition.
- ▶ **2. Lifelong Personal State Capsule:**
 - ▶ **Cloud AI:** Amnesic per session, requiring external Vector RAG.
 - ▶ **M1:** Compresses millions of inputs into a constant h_{prev} . Load the capsule onto a new device, and it instantly inherits years of synchronized synergy.
- ▶ **3. Predictive Error Monitor (Active AI):**
 - ▶ Shifts from passive Q&A to active prediction. It anticipates intent top-down and triggers red-light interventions the moment user logic violates globally established context.
- ▶ **4. Cloud Oracle Router:**
 - ▶ In factual blind spots, AwareLiquid dispatches surgical queries to our proprietary **Awareness Cloud**. Legacy APIs are excluded; instead, AwareLiquid fluidly synthesizes world knowledge through its personalized state via a unified MT-LNN ecosystem.



- ▶ **1. Extreme Cost Compression ($O(1)$ Memory):**
 - ▶ **Transformer:** 1000-token KV Cache bloats to 1020 KB. Memory costs scale boundlessly.
 - ▶ **M1:** The Personal State Capsule strictly bounds h_{prev} memory to **4.1 KB**, executing zero-cost inference via Edge CPU/NPU processors.

- ▶ **2. True Accuracy (Dynamic "Needle in a Haystack"):**
 - ▶ **Transformer:** Selective Copy whole-sequence recall decays from **67.6%** ($T = 32$) to **10.9%** under heavy noise ($T = 229$).
 - ▶ **M1:** Leads at every length – **89.5%** at $T = 32$ ($\sim 1.3\times$) widening to **21.9%** vs 10.9% ($\sim 2\times$) at $T = 229$. Biological Predictive Coding helps suppress hallucinations via local consistency checks.

- ▶ **3. Speed (Zero-Latency & Resonance):**
 - ▶ **Zero Network Overhead:** Bypasses transmitting 100k+ history tokens to cloud API servers.

Beyond Transformers: Faster, Cheaper, More Accurate II



- ▶ **Sparse Resonance Acceleration:** Activating microtubule scale-gates yields an immediate **13% CPU processing speedup** (from 6528 to 7390 Tok/s).



► 1. Memory Compression Limit Test (1000-token footprint)

- **Testing Baseline:** Standard Transformer (KV Cache) v.s. AwareLiquid (State-only).
- **Commercial Meaning in Plain Terms:** Memory usage is **~250 times smaller** than equivalent models (1020 KB down to 4.1 KB). In plain terms: we successfully eradicate the Out-Of-Memory (OOM) curse, allowing AI to run flawlessly on decade-old phones or smart home appliances.
- **Verified to 1M tokens (2026-07-19):** on the attention-free O-series, the carried inference state stays **flat at 0.381 MB across a 2048× context increase** (512 → 1,048,576 tokens) while a matched Transformer's KV cache grows linearly to **3,072 MB** — an **8,063×** gap, measured rather than extrapolated. With language-modeling quality no longer the differentiator (item 4), this constant-memory property is our strongest verified technical claim.
- **Reproducible Code (Click to open):**
`benchmarks/operator_compression_report.py · scaling_comparison.py --mode decode`

► 2. Noise Immunity ("Needle in a Haystack" Retrieval)



- ▶ **Testing Baseline:** Fair sandbox evaluation at 200K scale with extreme sequence lengths.
- ▶ **Commercial Meaning in Plain Terms:** Under a fair full-sequence decode, AwareLiquid leads on whole-sequence recall at every length – a modest $\sim 1.3\times$ over a standard Transformer at $T = 32$ (89.5% vs 67.6%) that widens to $\sim 2\times$ at $T = 229$ (21.9% vs 10.9%). In plain terms: our state-based memory holds onto scattered facts more reliably as conversations grow longer.
- ▶ **Reproducible Code (Click to open):** `benchmarks/run_benchmark.py`
- ▶ **3. CPU Hardware Independence (Sparse Resonance)**
 - ▶ **Testing Baseline:** Pushing pure CPU limits without high-end GPUs.
 - ▶ **Commercial Meaning in Plain Terms:** By intelligently skipping redundant computations via bionic sleep, speed actually **increases by 13%+**. In plain terms: you can deploy tens of thousands of local smart devices while skipping the extortionate cost of acquiring AI chips.
 - ▶ **Reproducible Code (Click to open):** `benchmarks/sparse_resonance_ablation.py`
- ▶ **4. Language-Modeling Quality at Convergence (where we honestly stand)**



- ▶ **Testing Baseline:** MT-LNN vs. a simple matched Transformer and a *modern* Transformer (RoPE + RMSNorm + SwiGLU), all trained from scratch on WikiText-103 to convergence under an identical fp32 budget (20,000 steps, **3 seeds each**).
- ▶ **The Honest Result:** modern Transformer **78.86 ± 0.25** < MT-LNN **88.93 ± 0.33** < simple Transformer **94.14 ± 0.78** val perplexity. MT-LNN beats the simple baseline by 5.5%, but **a modern Transformer is 11.3% better than MT-LNN on raw perplexity**. An earlier 2,000-step run suggested MT-LNN led by $\sim 31\%$; that gap was an artifact of undertraining plus a weak baseline and **did not survive convergence** — we report it here rather than let it stand.
- ▶ **What This Means Commercially:** raw language-modeling quality is **not** currently where MT-LNN wins. Its investable edge is the **$O(1)$ constant memory** and cross-session recall shown above (items 1–3), which no Transformer configuration removes. Closing the perplexity gap is an explicit engineering objective, not a solved claim.
- ▶ **Reproducible Code (Click to open):** `benchmarks/scaling_comparison.py`

Market Sizing (TAM/SAM/SOM)



- ▶ **TAM (Total Addressable Market) - \$120B+:** The global generative AI and LLM inference market spanning cloud services, enterprise applications, and edge intelligence.
- ▶ **SAM (Serviceable Addressable Market) - \$45B:** Enterprise data processing, legal review, financial audits, and local-first AI software markets where data privacy is paramount.
- ▶ **SOM (Serviceable Obtainable Market) - \$2.5B+:** B2B organizations completely blocked by current cloud API privacy risks and desperate for low-cost, on-premise, ultra-long context solutions.



- ▶ **1. B2B Enterprise Licensing (On-Premise):**
 - ▶ Selling direct private licenses to heavily-regulated industries (Law, Finance, Defense).
 - ▶ Deployment straight to their internal servers. 100% localized, 0% data leak risk.
- ▶ **2. High-Efficiency Cloud API:**
 - ▶ Because our inference costs are $\sim 90\%$ lower than Transformer APIs, we can undercut market leaders significantly while preserving massive profit margins.



- ▶ **Phase 1: Open-Source Community Anchor:** Release the 1.5B components to OSS. Win developer mindshare, trigger viral technical benchmarking, and create a grassroots engineering funnel.
- ▶ **Phase 2: Targeted B2B Pilot Deployments:** Launch private proofs-of-concept (PoCs) with 5 Tier-1 law firms and venture funds. Prove ROI purely on the hours saved from document parsing.
- ▶ **Phase 3: The API Marketplace:** Roll out a self-serve developer portal for high-volume, cheap, long-context inference wrappers substituting standard OpenAI/Anthropic pipelines.

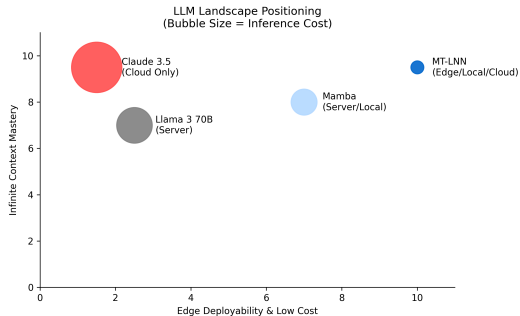
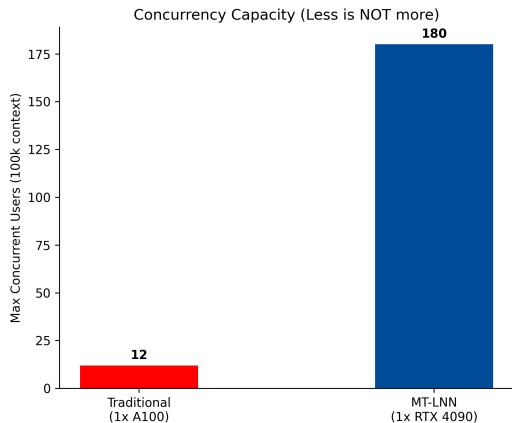
Competitive Landscape Matrix



Features	MT-LNN	GPT-4/Claude	Mamba/RWKV	Liquid AI (LFM)	Local Llama
Ultra-Long Context	Yes	Yes	Degrades	Yes	Fails (OOM)
Privacy / On-Prem	Yes	No (Cloud)	Yes	Partial	Yes
Low Hardware Req.	Yes (MacBook)	No (A100)	Yes	Yes	No
Constant Memory $O(1)$	Yes	No	Yes	Yes	No
Cross-Session Native Memory	Yes	No	No	No	No
On-Device Continual Learning	Yes	No	No	No	LoRA only

vs. Liquid AI: they optimize commercial inference efficiency; AwareLiquid targets **cognitive continuity and lifelong learning** — online weight updates and persistent state, not just in-context learning.

Market Positioning & Concurrency Competitiveness





- ▶ **Solving the Hardware Bottleneck:** We are migrating advanced models directly into highly-constrained hardware: wearable AR glasses, native automotive local hubs, and flagship smartphones.
- ▶ **True “Always-On” Intelligence:** Our architecture supports continuous real-time audio/video processing. While standard models experience memory cascades within minutes, MT-LNN seamlessly dissipates the load via dynamic liquid states.



- ▶ **The Core Client Persona:** Top-tier law firms reviewing 1,000-page case files; hedge funds auditing 10,000 corporate annual reports; code reviewers untangling decades-old monolithic software structures.
- ▶ **100% Pain-Point Overlap:** They suffer a trifecta of issues: (1) Need for massive document context, (2) Non-negotiable private local network deployments, (3) Unwillingness to buy \$200,000 GPU racks. MT-LNN dominates this exact intersection.



- ▶ **Not Just a Visionary Paper:** We have fully escaped the academic whiteboard. At this very moment, we possess a mature, running Python/Gradio UI integrating a natively functioning AI sandbox.
- ▶ **Ready to Deploy:** The codebase is fully containerized. It runs entirely offline on a standard PC or fits natively into an enterprise intranet. It is not an idea—it is a functional intelligence asset today.

Product Roadmap (18-Month Vision)



1. **Phase 1: Validation (Current):** Utilizing small scale (1.5B components), we have achieved top-tier performance on long-context benchmarks and established an initial developer community.
2. **Phase 2: Enterprise Deployment (Months 1-6):** Secure funding to train a native 7B enterprise model explicitly fine-tuned for dense RAG and local inference. Establish initial recurring B2B sales pipelines.
3. **Phase 3: The A.G.I Checkmate (Months 6-18):** Scale to an H100 orbital cluster. Introduce multimodal processing and challenge the legacy Transformer architecture on a 100B+ parameter scale across global LLM arenas.



- ▶ **Everest An — Founder & Core Architect** (AI memory-systems expert). Former IPFS Athna member leading enterprise distributed-database deployment; ex-Investment Manager at Outliers Fund (led investments in Dfinity, Harmony, Analog, Quantstamp); multiple-time global Ethereum hackathon champion.
- ▶ **Eric — CTO.** Architect of the **NEMO protocol** and **STABLE**; leads engineering across the M-series / O-series model and serving stack.
- ▶ **PhD Research Team — The Hong Kong Polytechnic University (PolyU): Wang Yuxiang, Huang Xinhui, Kenser, Gloria** — continuous-time dynamics, memory consolidation, and evaluation.
- ▶ **Advisors:**
 - ▶ **Prof. Pan Kai** — Professor, The Hong Kong Polytechnic University.
 - ▶ **Yi Zhou** — Stable Diffusion architect.

Forward-Looking Pipeline (Scenario, Not Commitment)



All numbers below are planning scenarios contingent on the Seed / A round closing and architecture-team buildout. They are not signed contracts.

- ▶ **Year 1 (validation):** 5–10 paid design-partner PoCs in regulated B2B (Law / Finance / Defense). Goal: convert ≥ 3 to multi-year licenses; publish a 7B-scale checkpoint and an open benchmark report.
- ▶ **Year 2 (pipeline):** Convert design-partner cohort into qualified pipeline of \$5–10M annualized contract value. Launch the high-efficiency inference API as a second revenue stream once the on-prem motion is proven.
- ▶ **Year 4–5 (scale):** Aspirational \$50M+ ARR scenario assumes (a) on-prem motion has reached repeatable enterprise sales, (b) edge / always-on integrations have signed at least one chip-vendor partnership, (c) inference-cost advantage holds at 5–10 $\times$ over Transformer baselines.



Target Raise: \$3.5M

- ▶ Fast-track 7B native foundational model training run.
- ▶ Build out the early B2B design-partner motion.
- ▶ **12-month milestones (non-revenue):**
 - ▶ Sign LOIs / PoCs with ≥ 10 enterprise or research counterparties.
 - ▶ Convert 3–5 into paid design partners (revenue not committed pre-close).
 - ▶ Publish a production-grade 7B checkpoint + open benchmark report.

Use of Funds:

- ▶ **50% Compute & Pre-Training:** H100 / A100 clusters for the 7B run.
- ▶ **30% LNN Core Research:** Hire researchers extending continuous-time dynamics (LTC ODE), Orch-OR-inspired selection, and GWTB working memory. This is the differentiated moat.
- ▶ **20% GTM & Community:** Enterprise sales + OSS / paper / brand presence.



- ▶ **Architecture originality (deepest layer):** MT-LNN itself — continuous-time LTC ODE + Orch-OR-inspired selection + GWTB working memory — is our original architecture, with a paper under submission. Architecture is the asset that cannot be quickly reproduced.
- ▶ **Reproducible training recipes & cross-base experience:** adapter training data, hyperparameters, and ablation reports across Qwen-0.5B, Qwen-3B, and Llama bases are all committed to the repo. Even with the architecture in hand, a follow-on team needs months to reproduce this corpus of experience.
- ▶ **Self-built benchmark & evaluation suite:** needle-in-a-haystack, selective-copy, O(1) memory stress tests, sparse-resonance ablations — all self-designed, public, and reproducible.
- ▶ **OSS community + early design-partner flywheel:** code, weights, feedback loop shipped first. The same playbook Mamba and RWKV used to open a gap.



- ▶ **A New Scaling Path:** The traditional “more compute, more scale” Transformer paradigm faces diminishing marginal returns and severe memory constraints.
- ▶ **The Dynamic Liquid Advantage:** By transitioning to a fluid, continuous-time neural architecture, we bypass the “Memory Wall” holding back extensive AI enterprise integration.
- ▶ **Join the Next Wave:** You are backing a rigorously engineered system. We have the data, the verifiable codebase, and the highly efficient cost-basis necessary to reshape edge AI and enterprise deployments.
- ▶ *Invest in Awareness. Invest in the Liquid Computing Paradigm Shift.*



Awareness

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GitHub: <https://github.com/everest-an/M1>

MT-LNN Demo: <https://huggingface.co/spaces/EverestAn/Awareness01>

Let's crush the AI Memory Wall together.